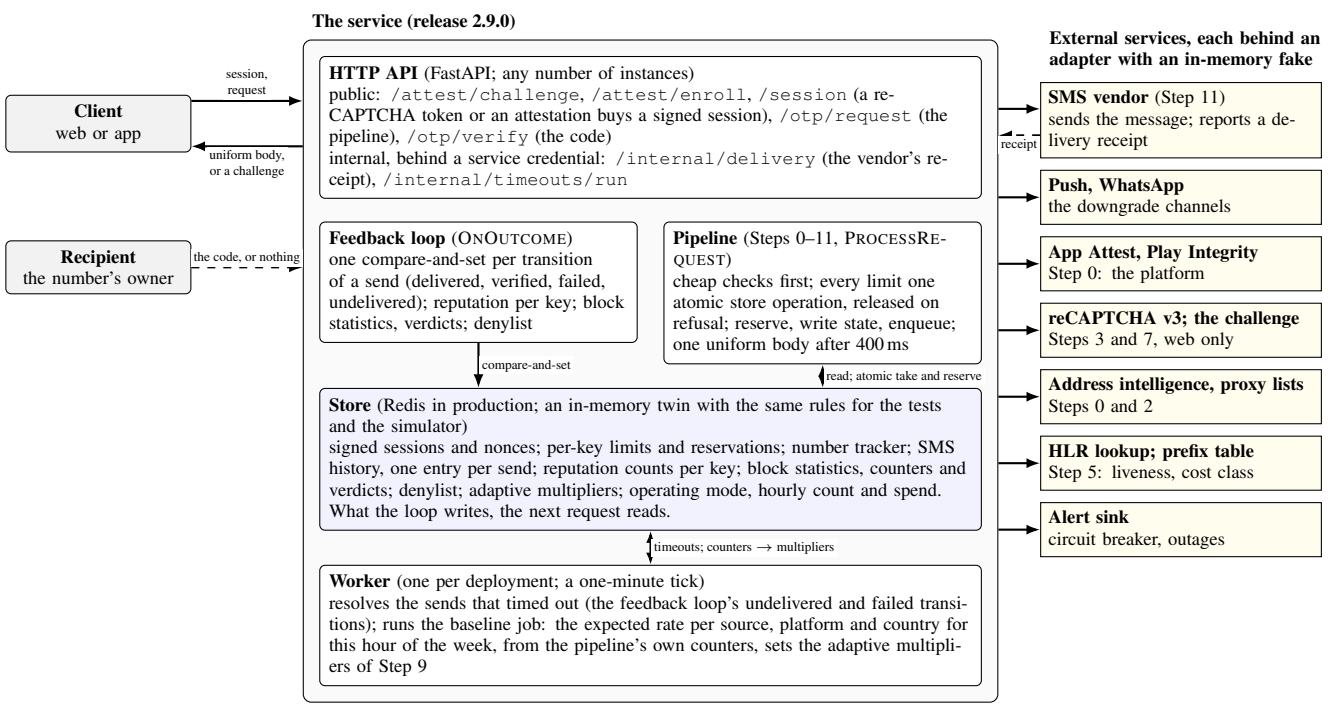
The service (release 2.9.0)
External services, each behind an adapter with an in-memory fake
Client
web or app
session, request
uniform body, or a challenge
Recipient
the number's owner
the code, or nothing
HTTP API (FastAPI; any number of instances)
public: /attest/challenge, /attest/enroll, /session (a reCAPTCHA token or an attestation buys a signed session), /otp/request (the pipeline), /otp/verify (the code)
internal, behind a service credential: /internal/delivery (the vendor's receipt), /internal/timeouts/run
Feedback loop (OnOutcome)
one compare-and-set per transition of a send (delivered, verified, failed, undelivered); reputation per key; block statistics, verdicts; denylist
Pipeline (Steps 0–11, ProcessRequest)
cheap checks first; every limit one atomic store operation, released on refusal; reserve, write state, enqueue; one uniform body after 400 ms
compare-and-set
read; atomic take and reserve
Store (Redis in production; an in-memory twin with the same rules for the tests and the simulator)
signed sessions and nonces; per-key limits and reservations; number tracker; SMS history, one entry per send; reputation counts per key; block statistics, counters and verdicts; denylist; adaptive multipliers; operating mode, hourly count and spend. What the loop writes, the next request reads.
timeouts; counters → multipliers
Worker (one per deployment; a one-minute tick)
resolves the sends that timed out (the feedback loop's undelivered and failed transitions); runs the baseline job: the expected rate per source, platform and country for this hour of the week, from the pipeline's own counters, sets the adaptive multipliers of Step 9
SMS vendor (Step 11)
sends the message; reports a delivery receipt
receipt
Push, WhatsApp
the downgrade channels
App Attest, Play Integrity
Step 0: the platform
reCAPTCHA v3; the challenge
Steps 3 and 7, web only
Address intelligence, proxy lists
Steps 0 and 2
HLR lookup; prefix table
Step 5: liveness, cost class
Alert sink
circuit breaker, outages